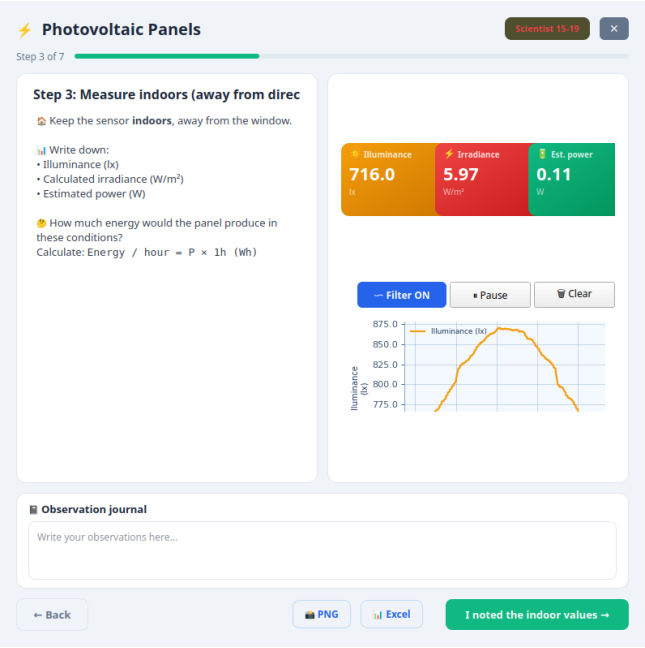
Solar Panels

A chain of assumptions, from lux to watts — and a protractor that proves why winters are cold

Level: Scientist — ages 15–19 (Romanian school years IX–XII) • **Time in class:** 30 min, one 50-minute lesson, or 3–4 lessons • **Sensors:** The light sensor (illuminance) and BME280 (air temperature)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Scientist — ages 15–19 (Romanian school years IX–XII). The activity has two halves that can be separated: the three-location series and the tilt series, which needs **the cosine**. The cosine of an acute angle is taught from **year VII**, so both halves work in any secondary-school year. The most direct entries are in **year VIII** (energy, power, efficiency, revisited here), **year IX** (trigonometry; general physical geography) and **year XII** (semiconductors, the *p*–*n* junction and the band gap).

Assumed knowledge

The cosine of an angle and reading a graph. Ratio and percentages. For the whole-day extension: the area under a curve by the trapezium rule. Calculus is **not** required.

Time in class

Short: 30 min (without the tilt series). Full: one 50-minute lesson. Project: three 50-minute lessons plus, optionally, a tracking day with readings every half hour.

Teacher preparation	25 min — **check the boxes' firmware**: with versions older than 2.2.5 the light sensor stops at about 3800 lx and the whole sunlit half comes out wrong (see the technical note); **cut the cardboard platforms yourself**, with a prop for each of 15°, 30°, 45°, 60° and 75°, or you will lose a quarter of the outdoor lesson to scissors; charge the laptops, because the tilt series runs away from a socket; wipe the sensor's window; and **walk the outdoor route**, looking for a spot with a clear view of the sun and no white wall nearby — light reflected off a bright facade visibly flattens the cosine curve.
Sensor settling time	**20 seconds** for light at each angle — the light sensor responds practically instantly. **But temperature needs 60–90 seconds**, and the efficiency correction depends on it; a box carried from the classroom into the yard reports the classroom's temperature for minutes afterwards.
Sensors and boxes	The light sensor for illuminance and BME280 for air temperature. One box per group of 3–4 pupils; with a single box the activity runs as a demonstration, and the roles in the adaptations box become compulsory so that everybody has something to do.

Three routes

Short	30 min: the introduction, the panel parameters and the three-location series — indoors, beside the window, in the sun — with the sun-to-classroom ratio said aloud. **No tilt series.**
Full	The 50-minute lesson: the short route plus the tilt series outdoors with a protractor, and a column-by-column comparison of $L(\theta)/L(0)$ against $\cos\theta$.
Project	Three 50-minute lessons: lesson 1 as above, lesson 2 the tilt series outdoors, lesson 3 the temperature correction, error analysis and the sizing calculation; optionally a tracking day with a reading every half hour.

1. Where this fits in the curriculum

A solar panel is simultaneously a physics problem, a geometry problem and an economic argument, and this activity touches all three without forcing anything. It suits:

- **Physics** — **year VIII** revisits exactly what is used here: energy, power and **efficiency** as the ratio of what comes out to what goes in. In **year XII**, the photoelectric effect (the photon, the energy threshold), semiconductors and the p–n junction explain why a silicon cell produces current when it is lit, and why there is a theoretical ceiling that cannot be passed. Mind the names: in a solar cell electrons are not "knocked out" of the material, as in the textbook's external photoelectric effect; they are lifted into the conduction band and separated by the junction's field (the photovoltaic effect).
- **Mathematics** — **trigonometry with a genuine purpose**. The cosine of the angle of incidence is not an exercise on a worksheet here; it is the number that decides how much power a roof produces. Direct proportionality, percentage change and curve fitting also appear.
- **Geography** — in **year IX**, general physical geography: solar elevation, latitude, the seasons, the angle of incidence and zonality. **Pupils who measure the cosine curve themselves with a protractor do not forget why winters are cold**, because they have watched the same energy spread over more ground.
- **Technology and practical applications** — sizing an installation, the payback period, orientation and tilt as design decisions, roof mounting.
- **Computing and ICT** — processing their own series in a spreadsheet: the ratio L/L_0, the graph overlaid on $\cos\theta$ and, for the tracking day, numerical integration.

The core skill is **modelling**. Pupils start from a raw reading in lux, pass it through a chain of

stated assumptions and arrive at a claim in watts. **Every link in the chain is questionable, and the questioning is the lesson.** At this level a pupil should finish able to say not only what the answer is but how much they trust it.

2. Time plan

The in-app wizard has ten steps — introduction, prediction, parameters, three locations, tilt series, temperature, **How sure are we?** and conclusion — and runs in half an hour without the tilt series. The physics only becomes convincing, however, when pupils go outside and change the angle *deliberately*, which needs a second lesson and reasonable weather. Choose the version that fits your timetable and your sky.

Route	Time	What happens
Short	**30 min**	The introduction and the panel parameters (9), the three-location series — indoors, beside the window, in the sun (15) — the temperature correction and the conclusions (6). The tilt series and the tracking day are skipped.
Lesson 1	50 min	Written predictions, the photovoltaic effect, the parameters entered in the app, the three-location series. First contact with the lux-to-watts conversion.
Lesson 2 (outdoors)	50 min	The tilt series with a protractor: illuminance against the angle of incidence, in 15° steps from 0° to 90°. The cosine law confronted with data.
Lesson 3	50 min	The temperature correction, error analysis and scale: from one small panel to a household and a country.
Tracking day (optional)	one day	A reading every 30–60 minutes from morning to late afternoon, then numerical integration for the day's energy in watt-hours.

The full project therefore occupies **three timetabled hours plus, optionally, one school day** of intermittent measurement. The tracking day works well alongside another lesson: a nominated pair leaves the room for ninety seconds every half hour, which costs almost nothing in teaching time and produces the best dataset of the whole unit.

Booking the weather

Plan the outdoor lesson **with a fallback already prepared**. A bright but overcast day still works for the tilt series: the values are smaller and the cosine relationship is weaker because much of the light is diffuse — which is an excellent discussion in itself, not a failure. What ruins the lesson is rain.

The safe fallback is **a bright desk lamp at a fixed distance in a darkened room**. It reproduces the cosine law almost perfectly and, frankly, more cleanly than the real sky, because it has no diffuse component. If you do both the outdoor version and the lamp version, the comparison between them is the best piece of work a pupil can produce on this activity.

3. Materials

Item	Per group	Notes
MagicBox with the light sensor and BME280	1	The light sensor stands in for the panel; the temperature sensor drives the efficiency correction.
Laptop with the app	1	USB or WiFi. A laptop that can be carried outdoors is a real advantage; otherwise a long USB cable through a ground-floor window.
Protractor, 180°	1	The largest you have. A printed protractor taped to card is easier to read outdoors than a small plastic one.
Stiff cardboard, A4 or larger	1 sheet	Cut as a tilting platform with a triangular prop, so the board sits at a set angle with the sensor lying flat on it.
Small photovoltaic panel and multimeter	1 (optional)	If the school owns one, it transforms the activity: pupils compare the predicted power with a measured open-circuit voltage or short-circuit current.
Ruler or tape measure	1	For the lamp distance and for the panel area.
Bright desk lamp	1	The wet-weather substitute for the sun, and the most precise way of controlling the intensity. Any lamp will do for the tilt series, because only ratios are compared; estimated watts, however, cannot be compared between a filament bulb and an LED — see the note on the conversion.
Compass, or a phone compass	1	For the azimuth the board faces outdoors.
Sticky tape and a marker	1 set	For fixing the sensor to the board and labelling the angles.
Clipboard	1	Outdoors, paper without a clipboard is paper in a hedge.

Preparation the day before. Charge the laptops fully, because the tilt series runs away from a socket. Cut the platforms and the props yourself, one for each angle, if you want the angles to be reliable and identical between groups. Wipe the sensor's window — **a fingerprint costs several thousand lux in direct sun**, silently. And walk the outdoor route, choosing a spot with a clear view of the sun and no white wall nearby.

4. Scientific background for the teacher

From photons to current

The photovoltaic effect, observed by Edmond Becquerel in 1839, is the appearance of a potential difference when light falls on a junction between two materials. In a silicon cell, a photon whose energy exceeds the band gap (about 1.1 eV for crystalline silicon) lifts an electron across the gap, leaving a hole; the built-in field of the p–n junction separates the pair and drives a current through the external circuit.

Two consequences matter here. First, photons with less energy than the band gap pass through without effect, and the surplus energy of high-energy photons is lost as heat — the underlying reason a single-junction silicon cell cannot exceed roughly 33% efficiency (the

Shockley–Queisser limit). Commercial modules reach 15–22%.
Second — and this is the link the whole activity rests on — because each absorbed photon yields at most one electron, the short-circuit current is very nearly **proportional** to the incident photon flux; the voltage grows only logarithmically, so the maximum power also follows the light almost in proportion. **That proportionality is the only reason a light sensor can stand in for a panel at all.** It is worth saying explicitly to the class: we are not measuring a panel, we are measuring the quantity a panel obeys — and the watts on the screen are **the output of a model**. They become a measurement only if you put a small panel beside the box and measure its current and voltage with a multimeter.

4.1 What the app actually computes

The app performs a deliberately simple four-step chain, which pupils should be able to recite:

$$E = \frac{L}{120} \qquad \text{irradiance in W/m}^2 \text{ from illuminance } L \text{ in lux} \qquad (1)$$

$$T_{\text{panel}} \approx T_{\text{air}} + 0.031\,E \qquad \text{estimated panel temperature} \qquad (2)$$

$$\eta = \eta_0\left[1 - 0.004\,(T_{\text{panel}} - 25)\right] \qquad \text{efficiency corrected for temperature} \qquad (3)$$

$$P = \eta \cdot A \cdot E \qquad \text{estimated power} \qquad (4)$$

where A is the panel area in square metres and η_0 the reference efficiency, both entered by the pupil on the wizard's parameter screen. The temperature correction applies **on both sides of 25 °C**: a cold panel is slightly *more* efficient than η_0, as in reality.

T_{air} **is the air temperature** measured by the box's BME280; the box has no panel. A panel in full sun is usually 20–30 °C warmer than the air, and the app estimates that excess with the NOCT rule: manufacturers quote a "nominal operating cell temperature" of about 45 °C for 20 °C air and 800 W/m^2, so $(45 - 20)/800 \approx 0.031$ °C per W/m^2 — some +25 °C in full sun and next to nothing indoors. **This too is a model**: wind, the way the panel is mounted and its colour change the real figure a lot. And a box left in the sun itself reads more than the air, because its case warms up; that is why the box stays in the shade for the temperature reading.

4.2 The lux-to-watts conversion is an approximation and should be challenged

Illuminance is a *photometric* quantity: it weights radiation by the sensitivity of the human eye, which peaks in the green at 555 nm and falls to zero in the infrared. Irradiance is a *radiometric* quantity and counts all wavelengths equally. There is therefore **no universal conversion factor** between them — it depends entirely on the spectrum of the source.

The divisor 120 corresponds to a luminous efficacy *of the radiation* of about 120 lumens per watt of light, a reasonable figure for sunlight (the direct beam alone is a little lower, around 100 lm/W): full sun at some 100 000 lux maps to roughly 830 W/m^2, close to the usual clear-sky value near 1000 W/m^2 at the ground. For other sources the factor is entirely different. A white LED lamp gives about 300 lm per watt *of emitted light* (the 100–200 lm/W on the box is per *electrical* watt and is not the quantity that matters here) and almost no infrared, so dividing by 120 **overestimates** the irradiance by about two and a half times. A filament bulb gives only 15–20 lm/W, because most of what it radiates is infrared the sensor cannot see, so dividing by 120 **underestimates** it six- to eightfold.

Ask the class what this implies about their indoor readings. A good group should conclude for themselves that indoor **comparisons** remain valid but that the **absolute watt figures indoors are not trustworthy**.

4.3 The cosine law: the heart of the tilt series

A parallel beam carrying irradiance I_0 perpendicular to its own direction of travel strikes a surface whose normal makes an angle θ with the beam. The same power is now spread over a larger area, by a factor $1/\cos\theta$, so the irradiance received per unit surface area is

$$I(\theta) = I_0 \cos\theta.$$

This is Lambert's cosine law. Its consequences are worth stating explicitly, because they are counter-intuitive:

- At $\theta = 30°$ the loss is only 13%. Alignment does not have to be perfect — which is why fixed roof panels are viable at all.
- At $\theta = 60°$ half the power is gone.
- At $\theta = 90°$ the received irradiance is zero, however strong the source.

The same law explains the seasons and the latitude dependence of climate: sunlight arrives at high latitudes at a large θ, spreading the same energy over more ground.

For a fixed installation, the optimum tilt from horizontal is approximately equal to the site latitude, facing the equator. For Romania (about 45° N) that is roughly a 35°–45° tilt facing south; the flattening towards lower angles allows for the extra summer yield when the sun is high.

4.4 Why hot panels perform worse

Rising temperature increases the intrinsic carrier concentration in the silicon, which reduces the open-circuit voltage of the cell by roughly 2 mV per degree per cell. The current rises very slightly, but the voltage dominates, so the power falls. The commonly quoted temperature coefficient is -0.3 to -0.5% per degree, and the app uses -0.4% per degree relative to the standard condition of 25 °C.

A dark roof-mounted panel easily reaches 60 °C in summer, which costs about 14% of its rated output — a striking result, since at exactly that moment the panel is receiving the most light of the year. This is why installers leave an air gap behind the modules.

Worked example for the board

A panel of area $A = 1.6$ m^2 and rated efficiency $\eta_0 = 19\%$ sits in the sun, with the module at 52 °C (30 °C air plus $0.031 \times 725 \approx 22$ °C, by the NOCT rule) and the sun 25° off the panel normal. The sensor, held **facing the sun squarely**, reads $L = 96\,000$ lx.

$$E = 96\,000/120 = 800 \text{ W/m}^2, \qquad E_{\text{panel}} = 800\cos 25° = 725 \text{ W/m}^2$$

$$\eta = 0.19\,[1 - 0.004(52 - 25)] = 0.19 \times 0.892 = 0.169$$

$$P = 0.169 \times 1.6 \times 725 \approx 196 \text{ W}$$

against a nameplate rating of about 304 W at standard test conditions (1000 W/m^2 on the panel, 25 °C). Ask pupils to account for the missing 108 W, taking the losses **one after another**, each applied to what is left after the previous one, so that nothing is counted twice:

$$304 \xrightarrow{800/1000} 243 \xrightarrow{\cos 25°} 220 \xrightarrow{0.892} 196 \text{ W}.$$

The weaker-than-standard sun costs about 61 W, the tilt about 23 W and the heat about 24 W — 108 W together, and **none of them is a fault in the panel**. Watch for one trap: if the sensor is taped *to* the panel, its reading already contains the cosine, and multiplying by $\cos 25°$ would count the tilt a second time.

5. Lesson flow

The in-app wizard has ten steps: a written-prediction step before the measurements, a step of its own for the tilt series and, before the conclusion, the **How sure are we?** step. Below is what the teacher does around the steps. The times are for the first 50-minute lesson; the tilt series, in square brackets, belongs to lesson 2.

1. **Written predictions, before anything is switched on (6 min).** Require three commitments in writing, with reasons: **(a)** how many times brighter is direct sunlight than the middle of the room — twice? ten times? a hundred? **(b)** if a panel is tilted 45° away from the sun, what fraction of its power remains? **(c)** does a panel work better on a hot day or on a cool bright one? Almost every class underestimates (a) dramatically, answers (b) as "half" instead of 71%, and gets (c) exactly backwards. Collect nothing; just insist that it is written and dated, because the value of this activity lies in the collision between these three answers and the data. The group also types its prediction into the app's prediction step; the conclusion at the end quotes it back next to the measured numbers.

2. **The introduction step (4 min).** Walk through the photovoltaic effect on the first screen — light creates electron–hole pairs and the field of the p–n junction separates them. Keep it brief: the real content is in the measurements.

3. **The panel parameters (4 min).** On the parameter screen, pupils set the area and the efficiency. Have them keep the defaults first (0.1 m^2 and 18%) so that every group is comparable, then deliberately change one value and watch the calculated power respond. Ask what kind of relationship they see between area and power. It is a free lesson in direct proportionality.

Technical note — the screen, the buttons and what is measured

The sensors. The light sensor for illuminance (the `LUX` column in the exported files) and BME280 for air temperature (the `T` column).
The light sensor's range depends on the firmware. With firmware 2.2.5 or newer, the box chooses its own sensitivity and measures up to about 120 000 lx, summer sun included. With older firmware the sensor **stops at about** 3800 **lx**: indoors is correct, but beside a bright window and outdoors the number sticks, and the sun-to-classroom ratio comes out tens of times too small. Reflash the boxes before the outdoor lesson (the flashing guide in the `firmware` folder). Above about 10 000 lx the absolute values are approximate (roughly ±15–20%): the relative comparisons (the tilt series, sun against shade) count, not the exact lux.
How it starts. The activity opens from the **Activities** page, from the **Solar Panels** card, with the **Start** button; the window that opens carries the same name.
The three cards: one measured, two calculated. At the indoor, window and sun steps the screen shows **Illuminance**, in `lx` (*Measured by the light sensor*), **Irradiance (model)**, in W/m^2 (*Calculated: illuminance ÷ 120*), and **Estimated power**, in W (*Calculated:* $P = \eta \times A \times E$), with a line underneath: *Measured: illuminance only.* The chart shows the illuminance over the last three minutes. If the reading sticks near 3800 lx, a yellow firmware warning appears. Only the illuminance is measured; the irradiance is the illuminance divided by 120, and the power comes from $P = \eta AE$. Ask pupils to say for themselves which number on the screen is measured and which is worked out — it is the cheapest exercise in scepticism in the whole guide, and the very question of the exit ticket.
The panel parameters. On the parameter step, two fields: **Panel area**, between 0.01 and 10 m^2, default 0.1, and **Panel efficiency**, between 1 and 30%, default 18. They feed straight into $P = \eta AE$ and can be changed at any time.
The steps and their buttons. The green button at the bottom right changes its label at every step, because it repeats what you have just finished: from **Got it, let's go!**, **My prediction is written** and **Parameters set**, through **I noted the indoor values**,

I compared the values, **I measured in the sun**, **I finished the angle series**, **I calculated the efficiency** and **Continue**, to **Finish**. The **Back** button goes back one step. At the measurement steps the **Capture** button keeps the reading; a step does not move on without at least one captured value.
The tilt-series step. It has the **Illuminance** card, a list **Angle θ to capture:** (0°, 15°, …, 90° and, at the end, **0° again (control)**), a table with the columns θ, L (lx), L/L_0 and $\cos\theta$, and a chart, **Light kept,** L/L_0, against **Angle θ (°)**: the **cos θ (law)** curve, the **your readings** points and the **control (0° again)** square. At every **Capture** the reading goes into the row of the chosen angle, and the list moves on to the next angle by itself.
The temperature step. The box now sits in the shade, so its light is no longer the sun's: the light and power cards go, and the screen shows **Air temperature**, **Panel temperature (model)** and **Corrected efficiency**. The panel temperature is $T_{\text{air}} + 0.031\,E$, with the irradiance E captured in the sun step ("Calculated from the air temperature and the sunlight you captured in the sun step"), and the corrected efficiency uses that temperature, not the air temperature directly. Pupils redo the chain by hand from the three cards.
The How sure are we? step. The app compares the control reading at 0° with the first one and writes the difference as a percentage: above a few per cent, the sky changed and the series is repeated. It also shows the largest gap between L/L_0 and $\cos\theta$ and reminds pupils that the factor 120 has no error bar.
How long you wait. For light, **20 seconds** at each position — just long enough for the hand that moved the board to leave the field. **For temperature, 60–90 seconds**, and the efficiency correction depends on it: a box carried from the classroom into the yard shows the classroom's temperature for minutes, and the efficiency displayed is then somebody else's.
Exporting the data. At the measurement steps, **PNG** and **Excel** appear; Excel saves the step's raw readings, unfiltered, with an `elapsed_s` column, plus the group's journal, and at **Finish** the app writes one Excel file for the whole exercise by itself. **There is no CSV button inside the activity**; a CSV file is taken from the **Data & Export** page: the **Excel** button in the box's row opens a save window in which CSV can be chosen as well. Save following the pattern `class_date_activity` — for example `11C_2026-09-14_panouri_solare` — and record **the angle, the time and the state of the sky** alongside the number, because without them the series cannot be interpreted afterwards.

4. **The indoor baseline (5 min).** Sensor at desk height, away from the window. Record illuminance, irradiance and estimated power. Expect between 150 and 500 lux and a power in the milliwatt range. Ask what a household would make of that figure.

5. **Beside the window, out of direct light (5 min).** The values typically rise by a factor of five to twenty. Pupils compute the *ratio* to the indoor value rather than just noting the number — the ratio is the transferable result.

6. **Direct sunlight (8 min).** Outdoors, or at an open sunlit window. Readings of 30 000 to 100 000 lux are normal — with the new firmware; if the number stops at about 3800, the box has old firmware and a yellow warning appears under the cards (see the technical note). This is where prediction (a) is settled. Have pupils state the full ratio, sun to classroom, out loud: values of 100 to 400 are routine and genuinely shock them.

7. [**Lesson 2: the tilt series, 25 min.**] The app has a step of its own for this series. With the sensor taped flat to the cardboard platform, take a reading with the board perpendicular to the sun's rays — $\theta = 0°$, found by **minimising the board's own shadow** — then at 15°, 30°, 45°, 60°, 75° and 90°. **Work quickly:** the sun moves, and thin cloud passing over invalidates comparisons between readings taken minutes apart. Pupils then compute L/L_0 and compare it with $\cos\theta$, column by column — the app fills in the same table alongside and plots the points over the cosine curve, so the hand calculation is checked on the spot. The last reading is the return to 0°, chosen from the list as **0° again (control)**.

8. **Temperature and efficiency (6 min).** On the temperature step, pupils read the air temperature, work out the panel temperature ($T_{\text{air}} + 0.031\,E$, with E from the sun step) and check it on the **Panel temperature (model)** card, then compute the corrected efficiency by hand and check it on the **Corrected efficiency** card. If you have a panel that has been sitting in the sun and an infrared thermometer, take its surface temperature and compare it with the estimate: **the gap between air and panel is large and surprising** — usually 20–30 °C — and the NOCT rule catches it only roughly. Keep the box in the shade for the temperature reading: in the sun its case warms up and reads more than the air.
9. **How sure are we? (5 min).** On this step the group reads the difference between the two readings at 0° and compares it with the gaps from $\cos\theta$. Say the rule out loud: a gap smaller than the control drift proves nothing against the law.
10. **Scale up and conclude (7 min).** Use the app's closing figures: about 150–200 kWh per year per square metre of panel in Romania, against a household consumption of 3000–4000 kWh per year, hence roughly twenty panels. Ask what area that is, whether it fits on a roof, and what happens at night. The storage question follows naturally and is the honest weak point of domestic solar.

Making the tracking day work

If you run the optional whole-day version, **give each pair a single time slot** rather than asking one group to do everything. Twelve readings taken by twelve pairs, pooled on a shared board or in a spreadsheet, produce a clean bell-shaped curve of irradiance against time. Pupils then estimate the day's energy by counting squares under the curve or applying the trapezium rule — their first genuine numerical integration, and one where the answer means something, in kilowatt-hours.

The investigation, in one table

The question. How much of the light falling on a panel is lost when the panel does not face the sun squarely?
The hypothesis. Written **before** any measuring: at 45° _____ of the power will remain, plus a reason. It is not changed afterwards. (The intuitive answer, "half", is wrong; the correct one is $\cos 45° = 0.71$.)
What I change. One thing only: the **angle** between the board's normal and the direction of the sun.
What I measure. Illuminance in lux, from the **Illuminance** card, after 20 seconds of stillness.
What I keep the same. The same box, the same place, the same person reading, the same background light, and **no shadow** — not of a head, a hand or the cardboard prop. Whoever holds the board stands so that the sun is behind the *sensor*, not behind them.
How many repeats. Two readings at each angle and, without fail, a **return to 0° at the end**. The whole series must be done in one continuous stretch of stable sky. **Tell the class in advance that a cloud will probably ruin one series**, so that it is expected rather than demoralising.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken at an angle read off the protractor, with the board held firmly, the sensor unshaded and the sky unchanged between the first and last reading. An artefact is anything else. **If the control value at the end differs from the one at the start by more than a few per cent, the series is discarded and repeated** — not "corrected".

6. Analysis after the activity

6.1 What the data should look like

Condition	Typical illuminance	Comment
Classroom, artificial light	150–500 lx	About 1–4 W/m^2; a 0.1 m^2 panel yields tens of milliwatts.
Beside a window, no direct sun	1000–5000 lx	Ten to twenty times the room value; pupils consistently underestimate this.
Overcast sky, outdoors	5000–20 000 lx	Still far brighter than any classroom.
Direct summer sun	60 000–100 000 lx	500–830 W/m^2 by the app's conversion. New firmware only; the old one stops at about 3800 lx.
Sensor tilted 45°	0.70–0.72 of the peak	Compare with $\cos 45° = 0.707$.
Sensor tilted 60°	0.48–0.55 of the peak	Compare with $\cos 60° = 0.500$.
Sensor tilted 90°	0.02–0.15 of the peak	**Not zero**, because of diffuse skylight and ground reflection.
Air temperature outdoors	15–35 °C	In full sun ($E \approx 800$ W/m^2) the estimated panel is about 25 °C warmer, 40–60 °C, so the corrected efficiency is about 16.9% down to 15.5% (with $\eta_0 = 18\%$). In the classroom, where E is small, the estimated panel is at about the air temperature and the efficiency comes out close to 18%, slightly above it below 25 °C.

The tilt data are the most rewarding. Plotting L/L_0 against θ and overlaying $\cos\theta$ usually gives close agreement out to about 60°, then a **systematic excess at large angles**. That excess is the diffuse component of skylight, which does not obey the cosine law because it does not arrive from one direction. **Naming that discrepancy correctly is a higher achievement than obtaining a tidy fit.**

6.2 Questions to ask afterwards

1. Your sensor measured lux, but a solar panel does not care about human vision. Explain why the app can still make a useful estimate of power, and name one situation where the estimate would be badly wrong.
2. At 90° the cosine law predicts zero, yet you measured a real value. Where did that light come from, and what does its size tell you about how much of the day's energy is diffuse?
3. A panel produces its greatest output in June and reaches its highest temperature in July. Using both the cosine law and the temperature coefficient, argue which month is likely to have the better actual yield.
4. Two identical houses fit identical panels, one facing south at 40° and one flat on a roof terrace. Predict the ratio of their annual outputs and state the assumptions in your calculation.
5. Your group and another group measured different values in the same sun within the same minute. Give three physical reasons that do *not* assume either group made a mistake.

6.3 Assessment ideas

- **Low stakes:** the pupil completes the three-location table and states the sun-to-classroom ratio in one sentence, then marks on the sheet where their original prediction sat.
- **Standard:** a graph of relative illuminance against tilt angle with the cosine curve overlaid, plus a paragraph explaining the agreement at small angles and the divergence at large ones.
- **Extension:** a costed design brief. Given a roof of stated area and orientation, a household demand of 3500 kWh per year and current prices, pupils size an installation, apply both the tilt and the temperature corrections, estimate the annual output and the payback period, and write a one-page recommendation defending their assumptions. Strong candidates **quantify** the uncertainty in the lux-to-watts conversion rather than ignoring it.

Common pitfalls and misconceptions

"Tilting the panel by half a right angle halves the power." The commonest numerical misconception. Pupils reason linearly; the cosine is not linear. At 45° about 71% survives, and it is worth making them predict before measuring so the error is theirs to correct.
Treating the calculated watts as measured watts. The app's power figure is a model output, not a measurement — only a small panel and a multimeter can check it. If pupils report it without qualification they have missed the point of the lesson. Insist on the full phrasing: *estimated power, assuming* 120 *lm/W and a panel efficiency of* 18%.
Believing hot and sunny is optimal. Prediction (c) is wrong for almost every class.
Confusing air temperature with panel temperature. The box measures the air; a panel in the sun is tens of degrees warmer. That is why the app does not put the air temperature straight into the correction but estimates the panel's with the NOCT rule — a model, not a measurement.
"Light knocks electrons out of the silicon." That is the picture of the external photoelectric effect. In a cell the electron stays in the material: it is lifted into the conduction band, and the field of the p–n junction sends it round the circuit. **The best day a panel ever gets is a cold, clear one with the sun high.**
Saturating or shading the sensor. A finger, the shadow of the cardboard prop, or the pupil's own head is the commonest cause of a nonsensical outdoor reading. A value stuck at about 3800 lx in full sun means old firmware, not a shadow. Establish that whoever holds the board stands with the sun behind the sensor, not behind themselves.
Comparing readings taken minutes apart under moving cloud. The whole tilt series is done in one continuous stretch of stable sky. If a cloud crosses, the series is discarded and repeated.
Confusing the tilt from horizontal with the angle of incidence. These are equal only when the sun is directly overhead. Outdoors the relevant angle is between the board's normal and the direction of the sun, found in practice by minimising the shadow the board casts on itself.
A dirty window. A fingerprint or dust costs several per cent, silently, on top of everything else. Clean it between groups.

Safety and classroom management

Nobody looks directly at the sun, and no pupil sights along the board towards it to judge the angle. **Teach the shadow-minimising method instead**: the board is perpendicular to the rays when its own shadow is smallest. **No lens, mirror or magnifier is brought to the outdoor lesson.**
Outdoors: count pupils out and in, define the working boundary before leaving the room, and keep the group away from car parks and roads. On a hot bright day, provide shade for the laptops and remind pupils about sun protection — a full tracking day means repeated

exposure at midday.
Indoors: filament lamps burn. Keep them clear of card, paper and cables, and let them cool before packing away. Cables trailing to a window are a trip hazard — tape them down.
If a real photovoltaic panel and multimeter are used: even a small panel in full sun produces a voltage that matters when several are in series; never short a large module, and never connect a panel to a multimeter set to a voltage range while measuring current. Glass-fronted panels are handled with two hands.
The box is not waterproof. It comes back indoors if it starts to rain, and it does not go on a wet surface.

Adaptations for pupils with special educational needs

The cosine table, printed. This is the main adaptation and it is not a concession: the $\cos\theta$ column arrives already filled in on the sheet, for every angle. The pupil does the division and the comparison — which are the skills being assessed — and looking up a cosine on a calculator, which is not, no longer eats their lesson.
The first row of the table, filled in advance. For pupils who find starting hardest, give the three-location table with the indoor row already written. Ask only for the sun-to-classroom ratio and a sentence about tilt.
Axes drawn in advance. For anyone whose whole lesson goes on constructing a graph, give a sheet with the axes drawn, scaled and labelled, and the $\cos\theta$ curve already sketched as a dotted line. The pupil lays their own points over it. The idea being learned is *the fit*, not the drawing of axes.
Three real roles for pupils with motor difficulties. Not being able to hold the board does not mean not taking part. The roles rotate at every angle:

- **the angle keeper** — reads the protractor and gives permission to measure; they have the right to stop a reading taken with the board wobbling, so they hold the validity of the whole series;
- **the screen reader** — says the illuminance aloud, rounded, and announces when the 20 seconds have passed;
- **the scribe** — writes the values down and works out L/L_0 as they arrive, which makes outliers visible on the spot rather than at home.

Holding the board is itself a suitable role for a pupil who needs a simple, clear physical task with a visible effect — it is essential and it builds confidence.
Sensory sensitivity and light. A few pupils find bright light unpleasant, and a few are photosensitive. Do not send them to the full-sun measurements and do not point the lamp at them; the scribe and screen reader roles are done in the shade and are just as valuable.

If things do not go to plan

It is raining, or overcast all week. Do not postpone. **A desk lamp at a fixed distance in a darkened room reproduces the cosine law more cleanly than the sky**, because it has no diffuse component. Run the tilt series there, and if a clear day appears later, repeat it outdoors: the two series side by side are the best possible piece of work on this activity, because the difference between them *is* the diffuse light.
It is overcast but you can go out. Run the series anyway and say in advance what will happen: a much gentler fall than $\cos\theta$. That is not a failure but the other half of the result.
You have one box. A demonstration, with the screen projected and you moving the board. Name an angle keeper, a screen reader and a scribe at every angle; the roles rotate, so every pupil has a real task. The table is filled in together and copied.
You have no large protractor. Cardboard props cut in advance, one per angle, are actually better than a small protractor held in the hand: the angle is identical for every

group and it does not wobble.
A cloud ruined the series and there is no time left. Keep the ruined series and use it: ask pupils to show *where* the sky changed, by comparing the control reading at the end with the one at the start. Recognising an invalid series is an assessable skill, not wasted time.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils write their predictions first, then receive the values row by row, compute L/L_0 and compare it with $\cos\theta$. The whole analysis — the fit at small angles, the excess at large ones, the diffuse component — is unchanged. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **National Renewable Energy Laboratory** — `https://www.nrel.gov` — search for solar resource maps and the PVWatts calculator, which lets pupils check their own annual yield estimate against a professional model for any location and tilt.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the photoelectric effect and energy forms simulations; the photoelectric one underpins the band-gap discussion pupils otherwise take on trust.
- **International Renewable Energy Agency** — `https://www.irena.org` — search for photovoltaic cost and capacity statistics, which supply the real figures for the payback-period task.
- **International Energy Agency** — `https://www.iea.org` — search for solar PV in the annual renewables report, for national and global context when pupils ask how much of the electricity mix this actually represents.
- **Institute of Physics** — `https://www.iop.org` — search for practical resources on light intensity and energy resources; useful for other classroom-tested variants of the cosine law demonstration.
- **NASA** — `https://www.nasa.gov` — search for solar irradiance and the solar constant, for the origin of the 1361 W/m^2 figure above the atmosphere and why only about 1000 W/m^2 reaches the ground.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *which of the numbers on the screen were measured and which were worked out?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Scientist level the three axes look like this:

- **Measurement practice** — the pupil finds $\theta = 0°$ by the minimum shadow, not by looking at the sun, respects the 20 seconds, avoids their own shadow and performs the control return at the end.
- **Data interpretation** — the pupil plots L/L_0 over $\cos\theta$, recognises the fit at small angles and *names* the excess at large angles as the diffuse component rather than treating it as error.
- **Scientific reasoning** — the pupil distinguishes the measured quantity from the calculated

one, states the 120 lm/W assumption and says under what source it fails, and explains why a cold clear day beats a hot one.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/solar_panels.pdf`. It has two pages — the three predictions and the three-location series on the first; the tilt series, the temperature correction and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The three predictions are written and dated at the start**, and the sheet leaves room beside each for the measured value — the collision between them is the lesson, and it has to be visible on the same page. **The $\cos\theta$ column is already printed** for every angle, so that the time goes into the comparison rather than into looking values up, and **the last row is the control return to** $0°$, without which the series cannot be declared sound. **There is a box for the state of the sky beside each series**, because a tilt series with no note of the sky cannot be interpreted a week later. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.